

Development of ROS based indoor autonomous flight Quadcopter Platform using Visual Odometry

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Abstract. Drones are used in various fields such as search, transportation, and agriculture. To control a drone in an indoor environment where GPS is unavailable, its pose and position data must be obtained from other devices, such as an onboard camera, an IMU sensor, or a motion-capture camera. In this paper, we propose a ROS-based indoor autonomous-flight drone platform that uses the visual-odometry algorithm of an onboard stereo camera. We also tested the drone flight-control algorithm in Gazebo simulation with PX4, an open-source flight-control firmware.

Keywords: Drone, Robot System, Robot Operating System

1. Introduction

Recent advances in drone technology have expanded their applications to transportation, agriculture, search, and other fields. In particular, drones are widely used at disaster sites, which are unstructured environments. Their high degree of freedom enables tasks such as searching for people in distress and detecting obstacles [1].

The Global Positioning System (GPS) is commonly used to track a drone's position during control. However, GPS is unavailable indoors, so an additional system is required to track the position of a remotely located drone.

Therefore, this study proposes a Robot Operating System (ROS)-based autonomous drone platform that uses visual odometry to track its indoor position. Gazebo simulation from the open-source PX4 package also enables testing of flight algorithms in various scenarios.

2. System Configuration

2.1 Hardware Configuration

The drone platform developed in this study is shown in Figure 1. It uses Holybro's open-source Pixhawk 6C as the flight controller, NVIDIA's Jetson Nano Developer Kit 4GB as the mission computer, and Stereolabs' ZED 2i stereo camera for visual odometry.

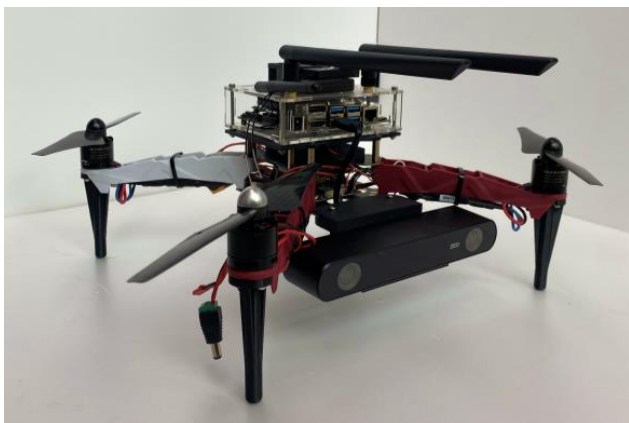


Figure 1. Indoor autonomous-flight quadcopter drone.

The flight controller and mission computer communicate over a serial connection through the mission computer's GPIO pins. The ZED 2i camera connects to the mission

computer through USB 3.0. A telemetry module is also attached to the flight controller to enable communication between the ground computer and the drone.

2.2 Software Configuration

The drone flight-control system uses the open-source PX4 v1.13.1. The mission computer runs NVIDIA's JetPack 4.6.1, based on Ubuntu 18.04. The ZED 2i firmware uses ZED SDK 4.0 provided by Stereolabs.

3. Autonomous-Flight Drone Platform

3.1 Visual Odometry

For autonomous operation, the proposed drone flies in Offboard mode, in which the mission computer or a ground computer can control it without a remote controller. To activate this mode on the flight controller, the drone's current position must be estimated in real time.

The ZED 2i stereo camera's visual odometry estimates the drone's position and orientation. Visual odometry is an algorithm that estimates camera motion in real time from consecutive images. It extracts feature points from consecutive images and compares them to calculate the amount of motion [2]. This allows the drone's position and orientation to be estimated relative to its initial state.

3.2 Robot Operating System

ROS is used to run the autonomous-flight algorithm on the mission computer, transmit commands to the flight controller, and exchange data between devices. The ROS1 package MAVROS transmits flight commands from the mission computer to the flight controller. MAVROS enables MAVLink, the message protocol used for drone communication, and delivers commands appropriate for the flight controller. Figure 2 shows the relationships between ROS nodes in a system configured to control the drone with the keyboard arrow keys.

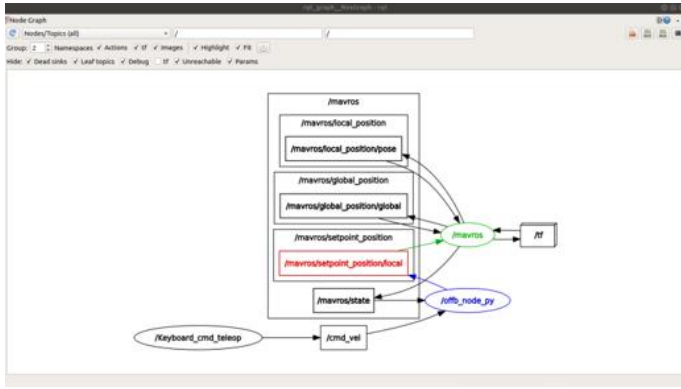


Figure 2. rqt graph.

The ZED 2i stereo camera's ROS package runs visual odometry. It publishes camera position and orientation messages on the Odom topic, allowing the mission computer to collect the data. A coordinate transformation aligns these data with the flight controller's coordinate frame, and the transformed position data are transmitted to the flight controller. The flight controller uses the received data to estimate position during flight. Figure 3 shows the position and orientation estimated by running the ZED 2i visual odometry.

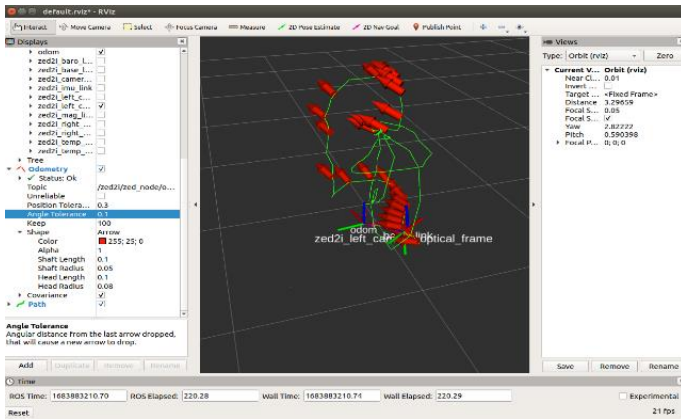


Figure 3. RViz.

3.2 Simulation

Flight algorithms must be tested in various scenarios to improve drone-flight stability. Gazebo simulation can test ROS control algorithms. Combined with PX4 firmware containing the control algorithm of the flight controller used in this study, it enables testing in an environment similar to that of the actual drone and can improve reliability and safety. Figure 4 shows the simulation environment used to test the mission computer's control algorithm through ROS commands.

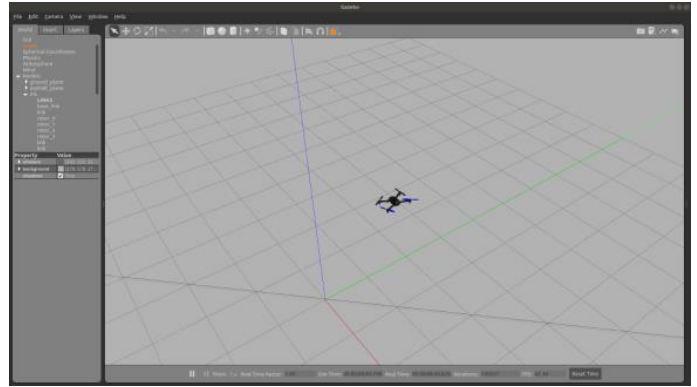


Figure 4. PX4-autopilot Gazebo.

4. Conclusion

This study developed an indoor autonomous-flight drone platform and confirmed through simulation that autonomous indoor flight is possible. The platform is expected to enable position estimation and position-based autonomous flight in indoor disaster environments where GPS is unavailable.

References

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